Bi–Al capacitor element with a bifilar coil

The project's first experiment: Q at resonance against stored energy per volume, with every joule traced to its port — CLAUDE.md §3e, milestone M5a

kronspice simulation report · 2026-11-07 · run 2, seed 1, 5 weighted-sum runs × population 40 × 30 generations per variant

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Summary

A bismuth–aluminium capacitor element and a bifilar coil (Tesla, US 512,340) are connected in parallel and left self-driven: the element is the only energy source, and the energy ledger must trace every joule stored or dissipated to the element's declared port. Evolution trades the tank's quality factor Q at resonance against the energy stored per volume, for three element variants: dry (Seebeck thermocouple, thermal port), wet (Al|electrolyte|Bi galvanic cell, chemical port) and thermopile (n Bi–Al pairs in series, thermal port).

- **Dry: microscopic storage.** Best energy per volume 1.254 µJ/m³ (analytic ½·ε·(EMF/gap)² bound 1.255 µJ/m³); highest Q 1,050 at 311.7 kHz with 5.4 pJ/m³. The 7.55 mV Seebeck EMF only charges the plates.
- **Wet: up to 89.3 J/m³,** mostly ½·L·I² from a 4.079 A steady cell current that a real aluminium electrode would not sustain (passivation): an upper bound until a cell is measured.
- **Thermopile: 31.25 J/m³ with 5,000 pairs** (EMF 37.75 V, EMF² scaling), but holding ΔT = 100 K across the legs conducts 47.83 W of heat to store 1.4 mJ.
- **Bifilar against ordinary:** the bifilar winding raises self-capacitance by about N²/4 and lowers the resonance (on 13 of 59 dry designs by more than 0.1 %), but leaves the stored energy unchanged and raises Q on no dry design.
- **The ledger closes:** worst per-point residual 1.6×10^{-14} of peak power, worst integration mismatch 7.9×10^{-4} of the energy taken in, model-supplied energy 0 J.
- **The coil model is superseded for prediction** (Knight 2016): the Q and resonance values must be re-run with the helical-line model before they are relied on.

Background

Tesla's US 512,340 winds two conductors side by side so that adjacent turns differ by half the coil voltage rather than V/N, which raises the coil's built-in capacitance; at 1,000 turns the idealized turn-to-turn energy ratio is (N/2)² = 250,000 (reproduced by TestTeslaComparison). This experiment asks what such a coil does when the tank is charged only by a Bi–Al element, and keeps the Tesla comparison separate from the realistic results, as §3e requires.

The element is also the bench for the temporal-field hypothesis **H4** (Jack 2003, em/temporal.go), which predicts, under rapid heating with an imposed anisotropy, an EMF from the time variation of temperature set by the Thomson coefficients and conductivities of Bi and Al. **Standard prediction:** the Seebeck EMF from the instantaneous temperature difference only, which is what this example computes. The difference is the measurement target; H4 itself is not modelled here and remains *untested*.

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The model

variant	element model	energy enters through
dry	Bi and Al plates with a solid dielectric; the Bi–Al leads form a thermocouple, EMF ($S_{Al} - S_{Bi}$)·ΔT = 75.5 µV/K·ΔT in series with the plate capacitance	thermal port
thermopile	the dry element charged by n Bi–Al pairs in series (n·75.5 µV/K·ΔT); genes add n (1–5000) and each leg's diameter and length; legs add series resistance, volume and a heat leak n·(k_{Bi} + k_{Al})·A/ℓ·ΔT; legs shorter than 5 diameters are rejected	thermal port
wet	Al|electrolyte|Bi galvanic cell (≤ 1.970 V), electrolyte resistance, Randles electrodes (double layer || charge transfer), geometric plate capacitance; plate capacity by Faraday's law	chemical port

- **Coil** (em/coil.go): Wheeler inductance; turn-to-turn capacitance from the two-wire formula with V/N per gap (ordinary) or V/2 (bifilar); skin effect at resonance; wire longer than λ/10 at resonance is rejected as not quasi-static.
- **Circuit:** the coil as series R and L with its self-capacitance across its terminals, in parallel with the element, between node a and ground. A 1 A AC probe current into a makes v(a) the tank impedance. For the switch-on transient the EMF steps on at t = 0 (idealized).
- **Resonance:** where the impedance phase falls through zero from inductive to capacitive; Q = (f_0/2)·|dφ/df| from the phase slope, f_0 fixed by bisection to 10^{-12}. A tank with no crossing is booked as no-measurement.
- **Genes:** turns 10–2000, coil diameter 5–150 mm, wire diameter 0.1–2 mm, pitch 1.05–3 wire diameters, insulation ε 2–6, plate side 5–150 mm, gap 10 µm–2 mm (wet 50 µm–5 mm), dielectric ε 2–100 (wet: set by the electrolyte); thermopile adds pairs 1–5000, leg diameter 50 µm–2 mm, leg length 1–50 mm.
- **Objectives:** Q at resonance, and energy per volume as the error $\log_{10}$(1 MJ/m³ / E_v). Five weighted-sum runs per variant trace the trade-off; non-dominated designs are reported. Every front design is re-simulated with an ordinary coil of the same geometry.

Declared assumptions

assumption	value	source
Bi–Al Seebeck coefficient	75.5 µV/K (S_{Al} +3.5, S_{Bi} −72 µV/K vs Pt)	Electronics Cooling (Nov 2006)
dry: plates above coil connections	ΔT = 100 K, EMF 7.55 mV	assumption
wet: cell EMF	1.97 V (Al^{3+}/Al −1.662 V, Bi^{3+}/Bi +0.308 V), the thermodynamic upper bound	CRC electrochemical series
wet: electrolyte	ρ = 0.21 Ω·m (sea water, S = 35, 20 °C), ε = 80	UNESCO 1981
wet: double layer, charge transfer	20 µF/cm², 10 Ω·cm² per electrode	Bard & Faulkner ch. 13 (typical)
plates	1 mm Bi + 1 mm Al	assumption
coil	copper; Wheeler L; two-wire turn-to-turn C; skin effect at f_0	Wheeler 1928; Tesla; Terman 1943
spectral line	Cu I 324.754 nm (air), 922.9 THz	NIST

Results (run 2)

Each variant: 5,710 evaluations. Dry and wet are bit-identical to run 1 (same seed); run 2 adds the thermopile.

variant	penalized	front	highest Q	highest energy per volume
dry	271	59 designs	39 turns, 150 mm, 311.7 kHz, Q 1,050, 5.4 pJ/m³	15 turns, 5 mm, 10 µm gap, 78.9 kHz, Q 0.75, 1.254 µJ/m³
wet	599	38 designs	23 turns, 141 mm, 11.16 kHz, Q 6.4, 4.7 mJ/m³	10 turns, 5.0 mm, 15.63 kHz, Q 2.4, 89.3 J/m³
thermopile	423	19 designs	3 pairs, 11 turns, 21.15 MHz, Q 1,250, 86 pJ/m³	5,000 pairs, 19 turns, 27.75 MHz, Q 43.3, 31.25 J/m³

Penalty reasons. Dry: coil too short for Wheeler 171, wire too thick 66, gap not small 34, no resonance 8, wire > λ/10 3. Wet: wire too thick 262, no resonance 216, coil too short 85, gap 46. Thermopile: legs shorter than 5 diameters 168, coil too short 147, wire > λ/10 59, gap 44, wire too thick 11.

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Extreme designs in detail (doc/experiments/bial_bifilar_run2.md)

design	L	self-C bifilar (ordinary)	steady state	switch-on transient ledger
dry, highest Q	225.5 µH	867.8 pF (2.282 pF; ratio 380.2)	no current; plates charge in 121.7 ps	in 613.7 pJ, dissipated 613.7 pJ, stored 8.216 fJ; residual 7.4×10^{-16}
dry, highest energy	1.264 µH	3.449 pF (61.32 fF; 56.25)	no current; τ 987.3 ns	in 1.645 nJ, dissipated 1.591 nJ, stored 55.69 pJ; residual 1.6×10^{-14}
wet, highest Q	82.66 µH	169.1 pF (1.279 pF; 132.2)	24.48 mA, 48.22 mW; plate lasts 8h13m	in 118.2 µJ, dissipated 91.19 µJ, stored 4.819 µJ; residual 8.2×10^{-16}
wet, highest energy	238.1 nH	14.56 pF (582.4 fF; 25)	4.079 A, 8.035 W; plate lasts 8h18m	in 13.1 mJ, dissipated 11.07 mJ, stored 792.1 µJ; residual 3×10^{-16}
thermopile, highest Q	1.469 µH	34.66 pF (1.146 pF; 30.25)	no current; heat leak 5.463 W	in 160.8 fJ, dissipated 159.8 fJ, stored 996.5 aJ; residual 2.5×10^{-16}
thermopile, highest energy	1.921 µH	17.12 pF (189.7 fF; 90.25)	no current; τ 33.02 ms; heat leak 47.83 W	in 395.7 nJ, dissipated 395.7 nJ, stored 32.86 pJ; residual 6.8×10^{-16}

Each ratio of bifilar to ordinary self-capacitance equals $N^2/4$ at that turn count. Transients are shorter than the plate charging time where stated, so they end with the plates partly charged. For the two thermopile extremes the ordinary-coil re-simulation is outside the lumped model (wire longer than λ/10 at the ordinary coil's higher resonance), which the report prints in place of the comparison.

What the fronts say

- **Dry.** The only store is the field in the plate gap, ½·ε·(EMF/gap)² diluted by the plates: 1.255 µJ/m³ by that formula against 1.254 µJ/m³ simulated for the best design. With a millivolt Seebeck EMF the energy per volume is necessarily microscopic. At the steady state no current flows (the plates block DC).
- **Wet.** The store is mostly ½·L·I² from the steady cell current, set by the assumed charge-transfer resistance: 4.079 A (96.9 mA/cm²) in the best design. The plate's charge lasts the same few hours (8 h 18 min) for every design because capacity and current both scale with plate area.
- **Thermopile.** 5,000 pairs raise the EMF to 37.75 V and the stored energy per volume from 1.25 µJ/m³ to 31 J/m³, the EMF² scaling ($5{,}000^2 = 2.5\times10^7$). Holding ΔT = 100 K conducts 47.83 W of heat (outside the electrical ledger) to store 1.4 mJ, and the plates charge through 16.86 kΩ with a 33 ms time constant.
- **Bifilar against ordinary.** At DC no voltage stands across the turn-to-turn capacitance, so the stored energy is the same for both windings. Lower resonance by more than 0.1 %: dry 13 of 59, wet 0 of 38, thermopile 4 of 19. Q raised by more than 0.1 %: dry 0, wet 0, thermopile 2.
- **Edges of the gene ranges.** The highest-energy optimum of every variant sits at the edge of some allowed ranges (e.g. dry: coil and wire diameter, insulation ε, gap, dielectric ε), so it is set by the chosen limits.

Ledger and validation

- No evaluated design failed the operating-point ledger check. Over the six switch-on transients the worst per-point residual is 1.6×10^{-14} of peak power, the worst integration mismatch 7.9×10^{-4} of the energy taken in, and model-supplied energy is 0 J: every joule comes from the element's declared port.
- Tests: TestTeslaComparison (250,000 at 1,000 turns), TestResonanceOfKnownTank (f_0, Q of a parallel RLC), TestDrySteadyState, TestWetSteadyState, TestThermopileScaling (EMF × n, stored × n²), TestOutsideValidityIsRejected (λ/10), TestQuickExperiment, TestExport.
- Resonances are also reported as the nearest whole-number subharmonic of the Cu I line; adjacent subharmonics are about 1/n apart with n ~ 10^7–10^{11}, so this is reported, not a constraint.
- The neutrino channel is listed as not modelled in this run.

Limits of the model

- **Superseded coil model.** Knight (2016, §6, §13 point 8) shows turn-to-turn capacitance theories have no predictive power for single-layer solenoids (Medhurst 1947 already found the model underestimates an ordinary coil's self-capacitance). The V/N model is kept to reproduce Tesla's idealized comparison; ordinary-coil self-capacitance and self-resonance belong to em/helix.go, and the strand coupling to the coupled-line model. The Q and resonance values above must be re-run before they are relied on (§3e).

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- The wet results are upper bounds: the charge-transfer resistance is assumed, and aluminium passivates well below the current densities found.
- ΔT = 100 K, the 1.97 V cell EMF and the plate thicknesses are assumptions; the EMF steps on instantly, while a real thermal or chemical source turns on over milliseconds or longer.

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Reproducing the results

```
go run ./examples/bial_bifilar                # about 1 min: report in runs/<time>-bial_bifilar/
go run ./examples/bial_bifilar -quick         # a few seconds
go run ./examples/bial_bifilar -deltaT 300 -emf-factor 0.6 -line Al
go run ./examples/bial_bifilar -netlists doc/experiments   # netlists + schematics of the best designs
go test ./examples/bial_bifilar
python examples/bial_bifilar/viz/make_report.py examples/bial_bifilar/bial_bifilar_report.pdf
```

The kept runs, their fronts (front_dry_run1.csv, front_wet_run1.csv, front_thermopile_run2.csv) and the netlists, LTspice files and equivalent-circuit drawings of the best designs are in doc/experiments/.

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